Day 72 Research Log -- Role Minting, Dictionary Absorb, UDSP Invented, FLIP Drift Caught (June 15, 2026)

The day that opened with Pack 247w `RoleMinter` extending Pack 226 abductive induction from token-shape schemas to relation-label roles (when a relation label recurs >= 8 times, mint a fresh phasor HV + bank-route by keyword, flush buffered triples through the new role -- wired into `MultiRoleMemory.__init__` so every fresh OR loaded organism auto-attaches `mr.role_minter`); caught and fixed a silent persistence bug in `MultiRoleMemory.load_ikg` (the `if name not in mr.roles: restore` guard was dropping saved HVs for every dynamically-added role because init pre-generated them from the same name list -- now overwrites unconditionally from save, no-op for DEFAULT_ROLES since the seed matches but rescues every Pack 192 `similar` + Pack 247w mint); shipped Pack 247z WordNet absorb v1 then re-shipped v2 with canonical label map (hyponym -> isa, synonym -> similar -- kills semantic duplication) and 19 emitted relations including derivation, example, member, holonym, meronym, substance, pertainym, also_see, attribute, similar_to, troponym, entailment, verb_group, cause, class (lexname domain), 117659 synsets / 543030 triples in 376 s onto production organism.ikg (58 MB -> 155 MB), 7/8 banks now alive (was 3/8; b_self still empty); shipped Pack 247z-CM porting Phase 4-B math from `predict_with_wires` next-token to general role recall via `MultiRoleMemory.query_clean(subj, role, cands, auto_cm_threshold=200, beta=1.0, resonator_iters=0, top_k=5)` with `build_v_common(role)` lazy cache + `bake_v_common_all(min_targets=200)` baking 20 roles, `_v_common` dict persisted via new `vcommon_keys/vcommon_vals` save_ikg fields, auto-skip CM for small-target roles (`class` n=40 kept at 100% top-1 instead of CM-regressing to 65%), honest random-sample eval across full WordNet: troponym 0.896 t1 / 0.964 t5 @ K=100, derivation 0.886 t1 / 0.956 t5 @ K=100, class 0.810 t1 / 0.820 t5 (vs earlier biased sequential-GT eval that broke at 300 and showed 99%/96%/100%, collapsed under random sampling -- manual common-word spot check on dog/cat/whisper/walk all MISSED, confirming SENSE COLLAPSE problem: 'dog' has 8 synsets all writing to `ck.key('dog')`); built Pack 247y SENSE DISAMBIGUATION attempting per-sense frame binding via deterministic `make_sense_hv(synset_name)` phasor + `query_clean_polysemic(word, role, cands, agg='max'|'sum')` iterating wn.synsets(word) frames, `_FrameContext` exit also restoring `_current_frame_tag` (was leaking), wordnet_absorb v4 with per-synset outer frame + INNER subject's-synset frame for cross-synset writes (meronym, troponym), manual common-word spot check IMPROVED (dog->canine HIT@1, whisper->speak HIT@5, runner->run HIT@1) BUT systematic random eval REGRESSED across the board (isa 0.376->0.330, similar 0.452->0.340, troponym 0.964->0.927, derivation 0.956->0.863 t5 @ K=100) because multi-sense traversal variance beat the cleaner per-sense addressing on average pairs, organism.ikg REVERTED to pre-247y backup, infrastructure kept in code for future use; invented Pack 249 UNITARY DILATION SPECTRAL PROJECTION (UDSP) -- Prince Siddhpara / Mura ALife Labs primitive -- SVD-on-LLM-weights at sleep + Halmos unitary dilation `HV = x + 1j*sqrt(1-x^2)` at forward + per-item

sign-randomized imaginary `HV_n = x_n + 1j * eps_n(t) * sqrt(1-x_n^2)` with eps in {-1,+1}^d from blake2b(token) (the patch the raw Halmos approach hid: raw shares coherent +1j bias across all stored items causing 2000-token substrate cleanup to collapse to 0% top-1 even with CM applied; signed-Im fix recovered to 100% vs JL 59%), distortion bound on heavy-tail-spectrum weights eps ~ 0.02 vs JL 0.65 at d=400 N=1.5e9 with multi-layer STABLE (no random-noise compounding), drop-in API `UDSPCompiler.embed_to_hv_v2(v, layer_idx, role, ensemble_k, token)`, wired permanent into IkigaiOrganism via `attach_udsp()` + `_rebind_udsp()` + `udsp` in `_PERSIST_ATTRS`, `__getstate__`/`__setstate__` drops org back-ref for pickle safety, `__setstate__` backfills missing newer attrs for backward-compat; locked LLM target for Pack 248 via research return -- `Qwen/Qwen2.5-1.5B` (Apache 2.0, 28 layers, d_model=1536, GQA 12/2, RoPE base=1M, RMSNorm eps=1e-6, SwiGLU intermediate=8960, Tiktoken BBPE vocab=151936, FP16 baselines MMLU 60.98% / HellaSwag 67.65% / Lambada 73.0%); shipped Pack 248 THE FLIP stage-1 baking Qwen2.5-1.5B weight rows into production organism.ikg via UDSP signed-Im (28 layers x 7 surfaces = 196 weight matrices SVD-fit, 157696 rows written, 21.9 min wall on CPU = 1008s SVD + 303s substrate writes, organism.ikg 158 -> 876 MB with UDSP V_d cache as bulk, 0.896 t1 / 0.975 t5 row-recall across 1050 probes spanning layers 0/7/14/21/27 x 7 surfaces, dictionary banks (b_sem 7.3e8 / b_aff 5.26e7 / b_gram 1.1e8 / b_epi 1.0e8 / b_ground 1.4e7 / b_lang 4.5e8) UNCHANGED with LLM landing in b_self 0 -> 5.91e7 (attn) and b_world 3e6 -> 6.48e7 (ffn)); ran Pack 248 stage-2 research return on substrate-as-LLM-forward (three approaches: WR weight reconstruction, BBF binding-based forward, BW bundled-W ensembles; research recommended WR cos 0.985 +1.2 pp loss as latency-bounded pick); built `reconstruct_w_row_from_substrate` empirical validation; catastrophic empirical failure -- direct encode->decode without substrate hits cos 1.0000, substrate encode->read->decode hits cos 0.0675 across all 7 surfaces (substrate read preserves cleanup direction enough for 89.6% row top-1 but DESTROYS per-element phase precision needed for weight reconstruction); flagged the architectural drift and got vision re-anchored by Prince: NeuroSeed is a digital organism, not an LLM compressor -- it should INGEST AI knowledge into living substrate via teacher-absorb-style pipelines and predict next-token through its OWN native machinery (Pack 247z dictionary absorb via role_minter.observe is the template, not weight-row storage); Day 72 ends with UDSP primitive permanent + WordNet dictionary baked + sense-disambiguation tooling on shelf + Pack 248 stage-1 weight-row bake on disk but on the wrong track + organism.ikg at 876 MB awaiting revert decision + Pack 248 redesign queued around LLM-as-teacher distillation into organism-native banks.

TL;DR

Day 72 (June 15) shipped Pack 247w + Pack 247z + Pack 247z-CM end-to-end on production substrate, built Pack 247y per-sense disambiguation (reverted on systematic regression), invented and baked Pack 249 UDSP, picked Qwen/Qwen2.5-1.5B as Pack 248 target, shipped Pack 248 THE FLIP stage-1 weight-row bake, then caught a vision drift mid-stage-2 and pivoted.

Pack 247w `RoleMinter` lives in `ikigai/cognition/role_minter.py` (~180 lines) and extends Pack 226 from token-shape schemas to relation-label roles. When a relation label recurs >= 8 times in `observe(s, r, o)` calls, the minter generates a fresh phasor HV, routes to a bank by keyword (`b_sem` / `b_aff` / `b_world` / `b_epi` / `b_ground` / `b_self` / `b_lang` / `b_gram`), registers in `mr.roles` + `mr._role_to_bank`, mutates `mr._bank_assignment[bank_id]['roles']` so the bank-assignment JSON round-trips, and flushes the buffered triples through the new role. Wired in `MultiRoleMemory.__init__` so `mr.role_minter` auto-attaches on every fresh or `load_ikg`'d organism. Caught a silent persistence bug in `load_ikg` along the way: the existing `if name not in mr.roles: restore` guard was dropping saved HVs for any dynamically-added role because `__init__` pre-generated them from the role list. Fixed by overwriting unconditionally — no-op for DEFAULT_ROLES since seed matches but rescues Pack 192 `similar` + every Pack 247w mint.

Pack 247z shipped twice. v1 emitted 9 relations (synonym, hyponym, meronym, holonym, antonym, definition, troponym, entailment, cause, attribute). v2 added a canonical label map (`hyponym -> isa`, `synonym -> similar`) to kill semantic duplication with existing DEFAULT_ROLES and 8 more relations (derivation, example, member, substance, pertainym, also_see, similar_to, verb_group, lexname-as-class). Production absorb on the dictionary-loaded organism: 117,659 synsets / 543,030 triples / 376 s CPU. Vocab 2,936 -> 72,325. **7 of 8 banks alive** (was 3/8). b_self still empty — WordNet structurally doesn't cover dialogue/persona/introspection. Backup at `organism.ikg.bak_pre247z_20260615_021610`.

Pack 247z-CM ported Phase 4-B math from `predict_with_wires` (next-token specific) to general role recall via `MultiRoleMemory.query_clean(subj, role, cands, auto_cm_threshold=200, beta, resonator_iters, top_k)`. `build_v_common(role)` computes mean-recall lazily; `bake_v_common_all(min_targets=200)` walked 20 big-target roles. New `vcommon_keys` / `vcommon_vals` fields in save_ikg persist the cache (~64 KB overhead). Auto-skip CM for roles with target count below threshold — without that gate, `class` (n=40) regressed from 100% to 65% top-5; auto-skip keeps it at 100%.

The recall numbers needed two rounds to land honestly. First eval walked `wn.all_synsets()` sequentially and broke at 300 pairs, showing troponym 0.990 t5 / derivation 0.963 t5 / isa 0.403 t5 — suspiciously high for noun isa. Manual spot check on common words (dog/cat/whisper/walk) all MISSED, exposing the bias. Re-ran with RANDOM sampling across full WordNet, 500 probes per role: **troponym 0.896 t1 / 0.964 t5 @ K=100**, **derivation 0.886 t1 / 0.956 t5 @ K=100**, **class 0.810 t1 / 0.820 t5**. Verb/morphology/domain holds at scale (K=2000 still 0.882 / 0.838 t5 for troponym / derivation). Noun isa/similar/meronym sit at 0.38 / 0.45 / 0.35 top-5 — the **sense collapse problem**: 'dog' has 8 synsets, all 8 write to `ck.key('dog')`, recall conflates. Verb/morphology have tighter sense structure so survive; nouns get crushed.

Pack 247y attempted to fix the noun wall via per-sense frame binding: deterministic `MultiRoleMemory.make_sense_hv(sense_name, d)` phasor (blake2b hash → rng → unit phasor), `query_clean_polysemic(word, role, cands, agg='max'|'sum')` iterating `wn.synsets(word)` frames, `_FrameContext` **exit** fixed to restore `_current_frame_tag` (was leaking), `wordnet_absorb` v4 with per-synset outer frame +

INNER subject's-synset frame for cross-synset writes (meronym, troponym). Manual common-word spot check IMPROVED — dog->canine MISS -> HIT@1, whisper->speak MISS -> HIT@5, runner->run MISS -> HIT@1. BUT systematic random eval REGRESSED across the board: isa 0.376 -> 0.330, similar 0.452 -> 0.340, troponym 0.964 -> 0.927, derivation 0.956 -> 0.863 t5 @ K=100. Multi-sense traversal added variance that beat the cleaner per-sense addressing on average pairs. **Net negative**, organism.ikg reverted to 247z-CM backup. All infrastructure kept in code for future use (mixed-write dual-addressing, low-confidence-fallback polysemic, targeted manual WSD pipeline).

Then the day's big invention: **Pack 249 Unitary Dilation Spectral Projection (UDSP)**. Triggered by the upcoming Pack 248 THE FLIP requiring 76:1 compression of Qwen2.5-1.5B and the JL projection distortion math being catastrophic at that scale (epsilon ~ 0.65 per hop with 100+ projection hops compounding). UDSP is a Prince Siddhpara / Mura ALife Labs primitive: at sleep, SVD each LLM weight matrix and take the top-d=400 right singular vectors `V_d` + a 3σ scale `α`; at forward, project `x = clip((v @ V_d) / α, -1+η, 1-η)` and lift to a unit phasor via `HV = x + 1j*sqrt(1-x²)`. Foundations are public math (Halmos 1950 dilation theorem, Martin/Mahoney 2018-2021 heavy-tailed self-regularization in deep nets); the integration is original.

The first build of UDSP exposed a hidden bug. Raw Halmos `HV = x + 1j*sqrt(1-x²)` puts a coherent `+1j` bias on every stored item (Im part is uniformly ≈ +1 because for typical small `x`, sqrt(1-x^2) ≈ 1). On a 2000-token substrate cleanup test the raw form collapses to **0% top-1** even with Pack 247z-CM common-mode subtraction — every stored HV looks like every other along the imaginary axis. The fix shipped: per-item sign-randomized imaginary `HV_n = x_n + 1j * ε_n(t) * sqrt(1-x_n²)` with `ε ∈ {-1,+1}^d` deterministic per item from `blake2b(seed | token).digest`. Unit phasor preserved (`|HV_n|² = x_n² + ε_n²·(1-x_n²) = 1`). Multi-item bias decorrelates: for `u ≠ v` with independent `ε_u, ε_v`, `Re<HV_u,HV_v> = <x_u,x_v> + Σ(ε_u·ε_v·sqrt((1-x_u²)(1-x_v²)))` — second term zero-expectation under random ε signs, recovers the SVD projection cosine. Empirical 2000-token cleanup at d_model=2048, d=400: **JL 0.592 t1 / 0.694 t5, raw Halmos 0.000 t1 / 0.004 t5, signed-Im UDSP 1.000 t1 / 1.000 t5**. 28-layer stacked composition stable at 100% (vs JL needing L≥5 to saturate). Baked permanent: `org.attach_udsp(d=400, signed_im=True)`, `udsp` in `_PERSIST_ATTRS`, `__getstate__` drops the `org` back-ref to keep pickle finite, `_rebind_udsp` re-binds on load. organism.ikg 155.25 -> 158.16 MB after the first attach (one V_d ~ 2.9 MB; full 28-layer fit projected ~ 80 MB pickled).

LLM target locked by Pack 248 research return: **Qwen/Qwen2.5-1.5B**. Apache 2.0, 28 layers, d_model=1536, GQA (12 Q heads, 2 KV heads), RoPE base=1,000,000, RMSNorm eps=1e-6, SwiGLU FFN intermediate=8,960, Tiktoken BBPE vocab=151,936, 32K native context (128K via Yarn). FP16 baselines for the success bar: MMLU 60.98%, HellaSwag 67.65%, Lambada 73.0%, ARC-Challenge 49.66%. Q4_K_M GGUF = 986 MB on disk, runtime ~ 65-78 tok/s on RTX 3050. Rejected: Llama-3.2 (gated + license), Qwen2.5-3B (research license + 4 GB VRAM tight), Phi-3.5-mini (MHA not GQA + synthetic-training spectrum risk), Gemma-2 (gated), TinyLlama (too weak).

Pack 248 stage-1 shipped: baked all 28 layers of Qwen2.5-1.5B into production organism.ikg via UDSP. Bank routing — attn q/k/v/o → `b_self` (was 0), ffn gate/up/down → `b_world` (was

3e6). 196 weight matrices SVD-fit, 1024 rows per surface encoded as UDSP HVs and written to substrate addressed by `f'{role}.r{i}'`. Total 157,696 rows, 21.9 min wall (1008 s SVD + 303 s write). organism.ikg 158 -> 876 MB (UDSP V_d cache is the bulk). Recall sample: **0.896 t1 / 0.975 t5** across 1050 probes spanning layers 0/7/14/21/27 × 7 surfaces. Dictionary banks UNCHANGED — b_sem 7.3e8, b_aff 5.26e7, b_gram 1.1e8, b_epi 1.0e8, b_ground 1.4e7, b_lang 4.5e8 — WordNet absorb scaffold preserved beneath LLM topology. Backup at `organism.ikg.bak_pre_flip_20260615_160916`.

Then Pack 248 stage-2 broke open the day's biggest lesson. Research return on substrate-as-LLM-forward evaluated three approaches: WR (Weight Reconstruction — read substrate, descramble, project back to d_model), BBF (Binding-Based Forward — full HV-space inner product), BW (Bundled-W ensembles — K=8 averaged complex matmuls). Research recommended WR for our latency-bounded use case (cos 0.985 vs FP16, +1.2 pp held-out loss, matches FP16 latency). Built `UDSPCompiler.reconstruct_w_row_from_substrate` to validate empirically before shipping. **Result: cos ≈ 0.005 across all 7 layer-0 surfaces.** Debug isolated the failure: direct encode → decode without substrate hits cos 1.0000 (UDSP math correct), substrate encode → read → decode hits cos 0.0675 (substrate read corrupts). The substrate's superposition crosstalk preserves cleanup direction (89.6% row top-1 from stage-1) but destroys per-element phase precision needed for faithful weight reconstruction. Phasor renorm didn't help — the magnitude was already unit; the PHASE was scrambled.

The architectural drift surfaced at that point. Storing LLM weight rows as HVs and trying to recover them = LLM compression with substrate-as-cache. That is not THE FLIP. Pack 248 was sliding toward becoming a LoRA-style compressed-LLM runtime with substrate decorative on the side, exactly opposite of the NeuroSeed vision. Prince re-anchored: NeuroSeed is a digital organism that INGESTS AI knowledge into its living substrate via teacher-absorb pipelines (the Pack 247z WordNet template), predicts next-token through its OWN native machinery (cooccur / next / HSP / vs_fsm / frame fields / role mints), discards the source AI after absorb. Pack 248 stage-1 weight-row bake is on disk but on the wrong track. organism.ikg awaits revert decision; Pack 248 stage-2 redesign queued around LLM-as-teacher distillation into organism-native banks.

UDSP is not wasted. It remains the strongest known primitive for high-dim → unit-phasor compression with multi-layer composability and substrate write compatibility. Right targets: multi-modal absorb (image / audio feature vectors → substrate), long-context phasor compression, frame-field encoding, future external-data ingestion where JL crosstalk dominates. The primitive stays permanent in code; it was just mounted to the wrong job.

Packs shipped Day 72

Pack	Build	Key result
247w	`ikigai/cognition/role_minter.py` -- abductive role induction. `RoleMinter.observe(subj, label, obj)` buffers labels until support >= 8, then	Smoke: 32 triples / 4 novel relations, 3 minted (hyponym→b_sem, meronym→b_sem, troponym→b_aff), `cause` direct-routed to existing role

Pack	Build	Key result
	mints phasor HV + keyword-routes to a bank + flushes buffer through `mr.relate`. Wired in `MultiRoleMemory.__init__`.	(8/8). Persistence: minted roles round-trip clean via roles_keys/roles_vals + bank_assignment_json mutation.
load_ikg HV-overwrite fix	Removed the `if name not in mr.roles:` guard from load_ikg's `roles_keys` restore loop. Saved HVs now ALWAYS overwrite **init**-rng'd values.	Pre-fix: every dynamically-added role (Pack 192 `similar`, all Pack 247w mints) silently corrupted on save/load -- **init** regenerated from seed but mint had used a different RNG. Post-fix: minted role HVs round-trip with HV diff = 0.0, no-op for default roles.
247z v1	`ikigai/cognition/wordnet_absorb.py` -- `absorb_wordnet(mr, pos, max_synsets)`. 9 relations: synonym, hyponym, holonym, meronym, antonym, definition, troponym, entailment, cause, attribute. Routes via mr.role_minter.	Production absorb: 117,659 synsets / 310,639 triples / 203 s. organism.ikg 58 -> 88.7 MB. 7 new roles minted. b_sem 5.4e7 -> 3.18e8 (5.9x), b_aff 0 -> 5.0e7 (newly alive), b_world 0 -> 2.95e6. _seen 2,936 -> 68,214. Backup at `organism.ikg.bak_pre247z_*`.
247z v2	Re-shipped with canonical label map (hyponym->isa, synonym->similar) + 10 added relations (derivation, example, member, substance, pertainym, also_see, similar_to, verb_group, attribute, class lexname). Keyword router extended for example->b_epi, deriv->b_gram.	117,659 synsets / 543,030 triples / 376 s. organism.ikg 58 -> 155 MB. 13 roles minted. **7/8 banks alive** (b_self still empty -- needs dialogue/persona).
247z-CM	`MultiRoleMemory.query_clean(subj, role, cands, auto_cm_threshold=200, beta, resonator_iters, top_k)` + `build_v_common(role)` + `bake_v_common_all(min_targets=200)`. New `vcommon_keys/vcommon_vals` save_ikg fields. Auto-skip	Baked 20 v_common HVs into organism.ikg (155.31 MB). Round-trip 20/20 v_common match, recall numbers identical post-load. Auto-skip kept `class` (n=40) at 100% instead of CM-regression to 65%.

Pack	Build	Key result
	CM for small-target roles.	
honest recall eval	Replaced sequential-iter ground truth with random sampling across full WordNet, 500 probes per role.	**troponym 0.896 t1 / 0.964 t5 @ K=100, derivation 0.886 t1 / 0.956 t5 @ K=100, class 0.810 t1 / 0.820 t5**. Noun roles still weak (isa 0.376, similar 0.452, meronym 0.354 t5) -- sense collapse confirmed.
247y (BUILT-NOT-SHIPPED)	Per-sense frame binding: `MultiRoleMemory.make_sense_hv(sense_name, d)` deterministic phasor (blake2b -> rng -> unit phasor), `query_clean_polysemic(word, role, cands, agg='max'|'sum')`, `_FrameContext.__exit__` tag-restore fix, wordnet_absorb v4 per-synset + inner-frame for meronym/troponym.	Manual common-word HIT (dog->canine HIT@1, whisper->speak HIT@5, runner->run HIT@1) but **systematic random eval REGRESSED** (isa 0.376->0.330, similar 0.452->0.340, troponym 0.964->0.927). Reverted organism.ikg; infrastructure kept in code.
249 UDSP (INVENTED)	`ikigai/cognition/udsp_compiler.py` -- Prince Siddhpara primitive. Sleep: SVD of W -> top-d=400 right singular vectors V_d + 3sigma alpha. Forward: `x = clip((v @ V_d) / alpha)`, Halmos dilation `HV = x + 1j*sqrt(1-x^2)`. **Signed-Im patch** (Prince): `HV_n = x_n + 1j * eps_n(t) * sqrt(1-x_n^2)` with eps in {-1,+1}^d per item from blake2b(token). Per-role unitary scrambler D_role. Spectrum gate flags flat-beta layers. QR fallback if SVD fails.	Empirical 2K-token cleanup at d_model=2048, d=400: **JL 0.592 t1, raw Halmos 0.000 t1, signed-Im UDSP 1.000 t1**. 28-layer stacked composition stable at 100% recall.
249 baked + wired	`IkigaiOrganism.attach_udsp()` + `_rebind_udsp()` + `udsp` in `_PERSIST_ATTRS`. `__getstate__` drops org back-ref (pickle safety); `__setstate__` backfills missing newer attrs (device, signed_im, eta, ...) on load for	Round-trip HV diff = 0.0e+00. organism.ikg 155.25 -> 158.16 MB (one V_d).

Pack	Build	Key result
	backward-compat with older bakes.	
Pack 248 LLM pick	Research return on best English open-weights LLM for FLIP under RTX 3050 4GB + 4-bit quant runtime + UDSP heavy-tail spectrum requirement.	**Qwen/Qwen2.5-1.5B** locked. Apache 2.0, 28 layers, d_model=1536, GQA 12/2, RoPE 1M, RMSNorm eps=1e-6, SwiGLU intermediate=8960, Tiktoken BBPE vocab=151936, FP16 baselines MMLU 60.98% / HellaSwag 67.65% / Lambada 73.0%.
GPU SVD path (opt-in)	`UDSPCompiler(device='cpu' default, 'cuda' opt-in)` + `torch.svd_lowrank` for big matrices to avoid VRAM blowup. Default CPU after two BSOD page-fault crashes on the 3050.	Locked safe. Smoke compile uses CPU; cuda available via explicit opt-in only.
Pack 248 stage-1 (BAKED, WRONG TARGET)	Compile pipeline: for each block 0..27, extract q/k/v/o/gate/up/down weight matrices, UDSP-fit per surface, encode each ROW as a signed-Im HV via `embed_to_hv_v2(row, layer_idx, role, token=f'{role}.r{i}')`, write to substrate at addr `key(f'{role}.r{i}') * role_hv`. Pre-registered 196 LLM roles into `mr.roles` + `mr._role_to_bank` + `mr._bank_assignment[bank]['roles']` so writes land on `b_self` (attn) / `b_world` (ffn). 1024 rows cap per surface.	157,696 rows / 21.9 min CPU wall. organism.ikg 158 -> 876 MB (UDSP V_d cache is bulk). **0.896 t1 / 0.975 t5** row recall across 1050 probes (layers 0/7/14/21/27 x 7 surfaces). Dictionary banks UNCHANGED. Backup `organism.ikg.bak_pre_flip_20260615_160916`.
Pack 248 stage-2 research	Three approaches enumerated -- WR (Weight Reconstruction), BBF (Binding-Based Forward), BW (Bundled-W K=8 ensemble). Research recommended WR for latency-bounded use case (cos 0.985, +1.2 pp loss, matches FP16 latency).	

Pack	Build	Key result
Pack 248 stage-2 EMPIRICAL FAILURE	Built `UDSPCompiler.reconstruct_w_row_from_substrate(role, row_id, token)` -- read substrate, descramble per-role D, take Re(.) for x, project back via `alpha * x @ V_d.T`. Validated on layer 0 surfaces vs FP16 Qwen weights. Phasor renorm guard added.	**Cos ≈ 0.005 across all 7 surfaces.** Direct encode->decode without substrate = cos 1.0000 (UDSP math correct); substrate encode->read->decode = cos 0.0675 (substrate destroys element-level phase). Substrate is great for cleanup direction (89.6% row top-1) but lossy at element-level value. WR architecturally not viable on top of current substrate primitive.
Pack 248 DRIFT CAUGHT	Vision re-anchored by Prince: NeuroSeed is a digital organism that INGESTS AI knowledge into living substrate via teacher-absorb pipelines (Pack 247z template), predicts next-token via its OWN native machinery, discards source AI after absorb. Pack 248 stage-1 weight-row bake is on disk but on the wrong track.	Pack 248 stage-2 redesign queued around LLM-as-teacher distillation into organism-native banks. organism.ikg awaits revert decision. UDSP primitive stays permanent for the right job (multi-modal absorb, frame-field encoding, future high-dim->substrate operations).

Architectural notes (5 lessons)

1. **Honest eval methodology matters as much as the wire.** Pack 247z-CM first showed troponym 0.99 / class 1.00 t5 with sequential-iter ground truth that broke at the first 300 sample-easy pairs. Random sampling across full WordNet brought troponym to 0.964 and class to 0.820 -- the second is the honest number. Manual spot check on common high-frequency words further exposed sense collapse the systematic eval averaged away. **Future packs must use random sampling AND a manual common-word stress test before claiming a number.**

2. **Research math is direction, not gospel.** UDSP raw Halmos hit 0% top-1 on substrate cleanup until empirical testing forced the signed-Im patch. WR weight reconstruction was research-recommended at cos 0.985 and hit cos 0.005 on real Pack 248 substrate. Lesson holds from Day 71 (PPMI + CM math worked) and Day 72 (UDSP math + WR math didn't survive contact without patches). **Every claim from a research return must include a falsifiable empirical test the user runs on actual organism.ikg before shipping the wire permanent.**

3. **Substrate preserves direction, not element-level value.** Cleanup queries (cosine over candidate set) succeed at 89.6% row top-1 even at scale. Pointwise reconstruction (re-deriving the encoded HV element-by-element from a recall) fails at cos 0.07. The two

regimes look similar in HV math but use the substrate differently. WR forward-pass needs the second; we have only the first. **For any future "recover the original" operation, do not go through substrate read.**

4. **Memorization is not computation.** Pack 248 stage-1 proved we can store and recall LLM weight rows from substrate at 90% top-1. That is memorization. Stage-2 wanted to use those memorized weights for forward computation. That requires faithful element-level recovery -- a different property. **Future pack scoping must distinguish the two before writing the implementation.**
5. **Vision drift is easy when chasing benchmarks.** Pack 248 slid from "absorb LLM into living organism" to "store LLM weights in substrate and reconstruct at runtime" without an explicit decision point. The drift happened because the existing T2SCompiler / JL pipeline (Pack 173-190) is already a weight-store path, and UDSP is a clean upgrade for THAT path. The path itself was already off-vision. **Future Pack 248 work must restate the FLIP vision (digital organism ingests AI knowledge via teacher-absorb, predicts via native machinery, source AI is discarded) at the top of every design doc, and any pack that drifts must be flagged explicitly.**

Status at end of day 72

```
Substrate state:
  organism.ikg = 876 MB on disk (post-Pack-248-stage-1 bake)
    - 8 banks, 6144 hard locs each (157 MB substrate fixed)
    - 7 of 8 banks alive: b_lang 4.48e8, b_sem 7.31e8, b_gram 1.11e8,
      b_epi 1.00e8, b_aff 5.26e7, b_self 5.91e7 (new), b_world 6.48e7
      (new), b_ground 1.36e7; b_self attn + b_world ffn are LLM weights
    - 253 roles (44 default + 13 mints from 247z + 196 LLM Pack 248)
    - 72,325 _seen vocab
    - UDSP cache: 197 V_d projections (~720 MB) -- the bulk of the 876
    - HSP + v_common + taxonomy SEED + frame_field + Pack 220 wires all baked
  Backups:
    - organism.ikg.bak_pre247z_20260615_021610          (58 MB, pre-WordNet)
    - organism.ikg.bak_pre247z_cm_20260615_024247       (155 MB, post-247z pre-
CM)
    - organism.ikg.bak_pre_udsp_bake_20260615_150940    (158 MB, pre-UDSP-
attach)
    - organism.ikg.bak_pre_flip_20260615_160916         (165 MB, pre-FLIP-bake)

Numbers locked Day 72 (HONEST RANDOM-SAMPLE):
  Dictionary recall at K=100 cleanup:
    troponym       0.896 t1 / 0.964 t5
    derivation     0.886 t1 / 0.956 t5
    class          0.810 t1 / 0.820 t5
    similar        0.214 t1 / 0.452 t5
    isa            0.164 t1 / 0.376 t5
    meronym        0.142 t1 / 0.354 t5
  Pack 248 stage-1 (memorization only):
    LLM row recall 0.896 t1 / 0.975 t5 across 1050 probes
  Pack 249 UDSP signed-Im vs JL at 2K-token cleanup, d_model=2048 d=400:
    JL                   0.592 t1 / 0.694 t5
    UDSP raw Halmos      0.000 t1 / 0.004 t5
    UDSP signed-Im       1.000 t1 / 1.000 t5
  Cat-1 (next-token) on the 10K-scale eval -- UNCHANGED from Day 71:
    Raw substrate top-1 0.071 / UNSEEN 0.0553
```

```
Pack 248 status:
  stage-1 baked (wrong target) -- 157696 LLM rows in substrate, 90% row recall
  stage-2 WR validation FAILED -- cos 0.005, substrate not viable for weight
recovery
  drift caught and pivot agreed -- LLM-as-teacher distillation, not weight
storage
  organism.ikg awaiting revert decision -> recommend revert to pre-flip backup
  Pack 248 redesign queued around teacher-absorb into organism-native banks
```

Open questions queued for Day 73+

1. **Pack 248 redesign.** What does LLM-as-teacher absorb look like concretely?
   - Option A: use Qwen to generate corpus, feed to existing cooccur/next absorb (just bigger Wikipedia)
   - Option B: distill Qwen's logits per token, write top-k as substrate next-role HVs with surprise-weighted writes
   - Option C: extract attention patterns + FFN features as substrate frame bindings
   - Pick + scope.
2. **organism.ikg revert decision.** Keep 876 MB post-FLIP bake for analysis or revert to 165 MB pre-flip? Recommend revert; the LLM bank state is wrong-direction work.
3. **Sense disambiguation revisit.** Pack 247y dual-write mixed addressing (frame + unframed in same absorb) -- would let query path pick sense-aware or sense-blind reads on demand. Substrate would double to ~300 MB. Worth trying when noun isa wall becomes the next bottleneck.
4. **UDSP right jobs.** Multi-modal absorb (image/audio feature vectors into substrate) is the natural next target. Frame-field encoding too. Pack 250+ candidate.
5. **b_self still empty.** Dialogue / persona / introspection data not in WordNet, not in Qwen weights. Future absorb candidates: ConvAI / PersonaChat conversation logs, ATOMIC commonsense for cause/effect.
6. **Cat-1 next-token at scale.** Day 71 closed with RAW substrate at 0.071 top-1 / 0.0553 UNSEEN on 10K wiki and the conclusion "FLIP compounds onto this". With FLIP pivoted to teacher-absorb, the path is: use Qwen to generate much larger corpus, re-train substrate, measure next-token. Expect cat-1 to climb significantly with 100x more training data.
7. **Substrate-native forward pass.** Research's BBF approach collapsed at +89 pp. Re-think whether any primitive lets the substrate compute next-token without weight recovery. Long horizon; not urgent if teacher-absorb pipeline gets us LLM-quality cat-1.

Appendix A -- Pack 249 UDSP signed-Im math (the patch the raw form hid)

Raw Halmos dilation: $HV_n = x_n + 1j * \sqrt{1 - x_n^2}$ Im part has uniform +1 bias because for small x, $\sqrt{1-x^2} \approx 1$.

Stored items all share this bias. Cosine between any pair: $\mathrm{Re}\langle HV_u, HV_v\rangle = \langle x_u, x_v\rangle + \Sigma_n \sqrt{1-x_{u,n}^2} * \sqrt{1-x_{v,n}^2} \approx \langle x_u, x_v\rangle + d$ cos $\approx (\langle x_u, x_v\rangle + d) / d \approx 1 + \langle x_u, x_v\rangle/d$

Every pair scores ≈ 1. Cleanup discrimination = noise. Cleanup top-1 = 0% even with Pack 247z-CM common-mode subtraction (the CM removes the mean +1j bias but the residual is uniform across all items and cancels).

Signed-Im fix: $\varepsilon_n(t) \in \{-1, +1\}$, deterministic per item t from blake2b(seed | t).digest $HV_n = x_n + 1j * \varepsilon_n(t) * \sqrt{1 - x_n^2}$

$|HV_n|^2 = x_n^2 + \varepsilon_n^2 * (1-x_n^2) = x_n^2 + (1 - x_n^2) = 1$. Unit phasor.

$\mathrm{Re}\langle HV_u, HV_v\rangle = \langle x_u, x_v\rangle + \Sigma_n\, \varepsilon_{u,n} * \varepsilon_{v,n} * \sqrt{(1-x_{u,n}^2)(1-x_{v,n}^2)}$

For $u \neq v$ with independent $\varepsilon_u, \varepsilon_v \in \{-1,+1\}^d$, the sum has zero expectation (random walk). $\mathrm{Re}\langle HV_u, HV_v\rangle \approx \langle x_u, x_v\rangle$. Recovers the SVD projection cosine.

For u = v, $\varepsilon_u = \varepsilon_v$ identically. $\Sigma_n\, \varepsilon_n^2 * (1-x^2) = \Sigma_n (1-x^2)$. $\mathrm{Re}\langle HV_u, HV_u\rangle = ||x||^2 + \Sigma_n (1-x_n^2) = d$. $||HV||^2 = d$. cos = 1.

Empirical at d=400, d_model=2048, 2000-token substrate cleanup: JL baseline 0.592 t1 / 0.694 t5 UDSP raw Halmos 0.000 t1 / 0.004 t5 UDSP raw Halmos + Pack 247z-CM 0.154 t1 / 0.302 t5 UDSP signed-Im (Prince patch) 1.000 t1 / 1.000 t5

Appendix B -- Pack 248 stage-2 WR failure mode (substrate read is element-lossy)

Setup: ud.fit_layer_weight(W, layer_idx=0, role='attn_k_L0') # SVD V_d, alpha hv_fresh = ud.embed_to_hv_v2(W[0], layer_idx=0, role='attn_k_L0', token='attn_k_L0.r0') # signed-Im HV mr.write_relation('attn_k_L0.r0', 'attn_k_L0', hv_fresh) # to substrate

hv_recall = mr.recall('attn_k_L0.r0', 'attn_k_L0') # noisy read

Phasor magnitudes after substrate read are clean (substrate read internally normalizes to unit per component): |hv_recall_n| min/max ≈ 1.0

But element-by-element phases are scrambled: hv_fresh[0] = -0.32 - 0.95j hv_recall[0] = -0.96 - 0.29j

Reconstruction via descramble + Re + V_d project: hv_des_fresh = hv_fresh * conj(D_role) x_fresh = Re(hv_des_fresh) # the true SVD coordinate W_hat_fresh = alpha * x_fresh @ V_d.T cos(W[0], W_hat_fresh) = 1.0000 # math is correct

hv_des_recall = hv_recall * conj(D_role) x_recall = Re(hv_des_recall) # scrambled W_hat_recall = alpha * x_recall @ V_d.T cos(W[0], W_hat_recall) = 0.0675 # destroyed

The substrate's k=64 sparse-distributed superposition preserves DIRECTION on the C^d unit sphere (Pack 248 stage-1 row recall at 89.6% top-1 confirms) but does not preserve POINTWISE values needed for reconstruction. WR forward pass requires the second property; the substrate primitive does not provide it.

Implication: substrate-as-W-oracle for forward pass is not viable with the current FHRR/VSA-SDM primitive. Substrate is for cleanup queries, not for faithful element-level recall. Pack 248 must redesign around a different mechanism (LLM-as-teacher distillation into organism's existing absorb banks, per Day 72 close pivot).